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PALAEOMAGNETIC PROPERTIES OF EOCAMBRIAN SEDIMENTS IN NORTHWESTERN SCOTLAND: IMPLICATIONS FOR WORLD-WIDE GLACIATION IN THE LATE PRECAMBRIAN

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ABSTRACT

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Samples from two regions and from a range of rock types in northwest Scotland have remanences that are stable to thermal and alternating magnetic field demagnetisation. Normal and reversed directions for different areas are also in closer agreement after correction for the different tectonic tilt in the different areas. The age of the remanence is pre-Devonian and most considerations indicate, but do not prove, an Eocambrian age. These data are therefore thought to substantiate previous evidence proposed for the low latitude of deposition of the Eocambrian tillites and associated thick limestones and dolomites. On this basis, it is not thought that the climatic changes indicated by the nature of the sediments can be attributed to rapid polar wandering during this period. The data do, however, indicate that the glaciation not only took place at sea level, but at latitudes which indicate that these glaciations at this time could represent a world-wide glacial event.

INTRODUCTION

The occurrence of glaciation towards the end of Precambrian times has been widely recognised in all continents, except Antarctica (Harland, 1964; Harland and Rudwick, 1964; Dunn et al., 1971; Crawford and Daily, 1971; Williams, 1974, 1975; and others). At least three distinct glacial episodes are recorded in several continents, particularly in Africa and Australia (Williams, 1975) the oldest is the Gnesjo glaciation some 960 million years ago, the Startian was some 770 million years ago, and the youngest, the Varangian, occurred some 615 million years ago. The duration of each of these episodes is even less certain than their age, but it is generally thought that the thickness of the glacial sequences indicate episodes of tens of millions of years duration. Each of the glacial sequences is interspersed with non-glacial

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(warmer) deposits in a comparable way as the ice sheet advances and retreats recorded in Late Cenozoic deposits. Mountain glaciations can, of course, occur at many times throughout the history of the Earth, but extensive ice sheets are rare. The formation of younger ice sheets, such as the Sakmarian (basal Permian) glaciation of the Gondwanan continents and the Quaternary ice sheet glaciation can be understood, in very general terms, in the context of specific palaeogeographic conditions existing at high latitudes (Tarling, 1978), but these Eocambrian glaciations are particularly intriguing as they appear to be world-wide in extent and do not appear to be associated with orogenic events (and hence are not mountain glaciations) and the interglacial deposits frequently suggest that they were formed in very low latitudes. The apparent global scale of the Varangian glaciation has been suggested as the direct cause of the development of skeletal life-forms that mark the beginning of Cambrian times, although this correlation is, in fact, dubious as there appears to have been a delay of some 20 million years or more between the cessation of the glaciation and the start of the Cambrian. Nonetheless, the problem of the nature and origin of this glaciation remains unresolved.

Three main origins have been proposed: (1) A world-wide synchronous glaciation such as could occur by a reduction in solar radiation reaching the Earth, possibly due to the solar system passing through an interstellar dust cloud or changes in the solar radiation "constant" (Harland, 1964; Dunn et al., 1971). (2) A "normal" high-latitude glaciation at a time of rapid polar motion so that the glacial debris are not, in fact, synchronous, but somewhat different in age in different areas (Crawford and Daily, 1971; Piper, 1973; McElhinny et al., 1974). (3) Climatic oddities arising because the Earth's axis of rotation lay close to the ecliptic so that there was, effectively, six months day and six months night, the effect of which would oscillate from excessive cold to excessive heat depending on "normal" precessions of the Earth's rotational axis (Williams, 1974, 1975). Clearly, the main test between such models is that of the degree of synchroneity between the different deposits, but such speculations can be reduced if it could be demonstrated that all the glaciations did, in fact, occur at high latitudes as this would, in itself, mean that the glaciations could not be coeval. Palaeoclimatic indicators of palaeolatitude are notoriously ambiguous, although the presence of thick carbonate sequences, with evidence of evaporites and the presence of stellate structures within some Eocambrian glaciation sequences, such as in northwestern Scotland, would imply low latitudes. Palaeolatitudes can, however, be determined quantitatively from palaeomagnetic data if the original magnetization of the sediments can be isolated and if the average geomagnetic field at that time was essentially that of an axial geocentric dipole field. Attempts have been made to evaluate the probable palaeolatitudes of several Late Precambrian glacial sequences by interpolation for this time within the existing polar wander paths, but such estimates suffer from the imprecision in the dating of both the polar wander curves themselves and of the glacial deposits. Palaeomagnetic observations directly on these glacial deposits, or intimately associated beds, are sparse although observations in Greenland

and Norway (Harland and Bidgood, 1959; Bidgood and Harland, 1961), North America (Morris, 1977), Africa and Australia (McWilliams, 1977; McWilliams and McElhinny, 1980) and Greenland and Scotland (Tarling, 1974) all seem to indicate low latitudes, although some of the Australia palaeolatitudes are as high as 60° (McWilliams, 1977). In order to extend the evidence for the low latitude origin of the Eocambrian glacial deposits in northwestern Scotland, a more extensive collection of samples has been made in three localities which deliberately included intimately associated thick sequences of limestones and dolostones which immediately preceded the deposition of the glacial tillites and mixtites.

THE PORT ASKAIG FORMATION IN NORTHWESTERN SCOTLAND AND PREVIOUS MAGNETIC STUDIES

In Britain, the Dalradian sequences of Islay (55.8°N6.1°W) and the Garvellachs (56.2°N5.8°W) have been studied in great detail (Spencer, 1971). Glacial tillites and mixtite are interbedded with dolomitic deposits of the Port Askaig Formation which is overlain by the Bonahaven Dolomite and underlain by the Islay Limestone (Spencer and Spencer, 1972). The presence of such thick limestones and dolomites suggest that low to intermediate latitude prevailed in these areas during most of the Dalradian. Tarling (1974) presented palaeomagnetic results from graded siltstones and some mixtites of Port Askaig Tillite. This study comprised some 57 samples from 9 sites on Garbh Eileach, the largest of the three Garvellach Islands; 54 samples came from a 6 m thick varved siltstone bed and 3 samples were from a mixtite bed. After thermal treatment and correction for tectonic tilt, two groups of directions were isolated, one with a shallow southerly direction (166, −12°, α_{95} = 6°) comprising 26 samples from 5 different sites, and a steeper northeasterly direction (59°, 57°, α_{95} = 10°) comprising only 4 samples from 3 sites in the varved siltstone bed. These directions correspond to pole positions at 39°S, 10°E and 17°S, 129°E, respectively. It was suggested that the main group represented the palaeomagnetic field at the time of deposition of the beds, whereas the second results represent the palaeomagnetic field sometime during the Ordovician, probably at the time of the Caledonian metamorphism. On that basis, the palaeolatitude at the time of deposition was estimated to be less than 10° and, at the time of the metamorphic event, it was some 40°. A major limitation of this study was the relatively small number of samples and rock types. Since the beds suffered low-grade metamorphism during the Caledonian orogeny (Spencer, 1971; Borradaile and Johnson, 1973), there were also the possibilities of re-magnetization and distortion of the magnetic minerals during the formation of the cleavage. Abouzakhm and Tarling (1975), working on the same sample collection, showed that the dynamic metamorphism did not apparently affect the palaeomagnetic directions of the rocks studied as the palaeomagnetic directions were unrelated to the petrofabric of the magnetites.

Re-examining the results given by Abouzakhm (1974) and Abouzakhm and Tarling (1975) shows that although the remanent direction and the magnetic susceptibility dependent orientations are unrelated, the magnetic fabric is not a primary depositional fabric, but is of strain-controlled origin. The maximum susceptibility (k_1) lies normal to the bedding plane and the minimum susceptibility (k_3) was close to the intersection of the bedding and cleavage planes (fig.2 in Abouzakhm and Tarling, 1975). Further computations of other magnetic anisotropy parameters (Fig.2) strongly support a strain-controlled fabric. For instance, the q-parameter is generally higher than 0.67, with values up to ~1.7; the angle V is also within the range of strain-controlled fabric and the susceptibility ellipsoid shape is predominantly prolate rather than oblate. The degree of anisotropy is, however, small. Borradaile and Spencer, in the discussion of Tarling's (1974) paper, suggested that there could have been steepening of the magnetic fabric relative to bedding but this would only result in an increase in palaeolatitude from 8° to 11°—15°. More recent evaluation of the Digico anisotropy delineator used to determine this fabric indicates that the instrument overestimates the actual fabric, thus further reducing any direct affect of the fabric on the remnant direction. These data were, however, still based on samples from essentially one small outcrop and were all from glacial beds.

GARB EILACH, GARVELLACH ISLANDS

The sampling on the Garvellachs has been extended by a further 178 samples from 30 sites (Fig.1) covering different lithologies including varves, dolomitic beds, grey, blue and pink limestones, mixites and dolomitic

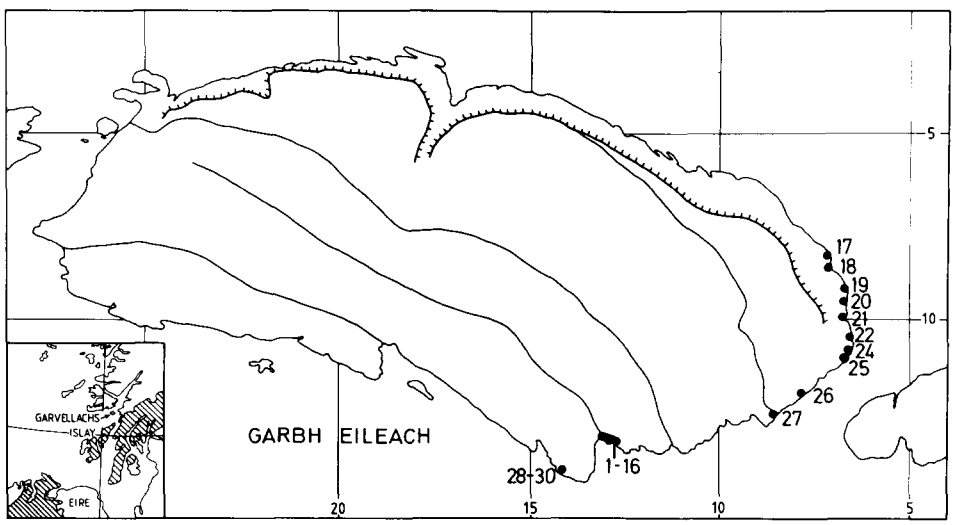


Fig.1. New sampling sites in the Garvellach Islands.

mixtites. Variation in intensity of initial remanence defined two clearly distinct groups; low values ($\leqslant$0.6 m A/m) and high values ($\geqslant$3 m A/m, with some as high as 130 m A/m). The directions showed relatively small within-site scatter, particularly those sites of high intensity, with most sites showing shallow reversely polarized directions. Two sites showed normal polarity (Fig.3 and Table I) and about 11 sites, all except 2 within the low intensity

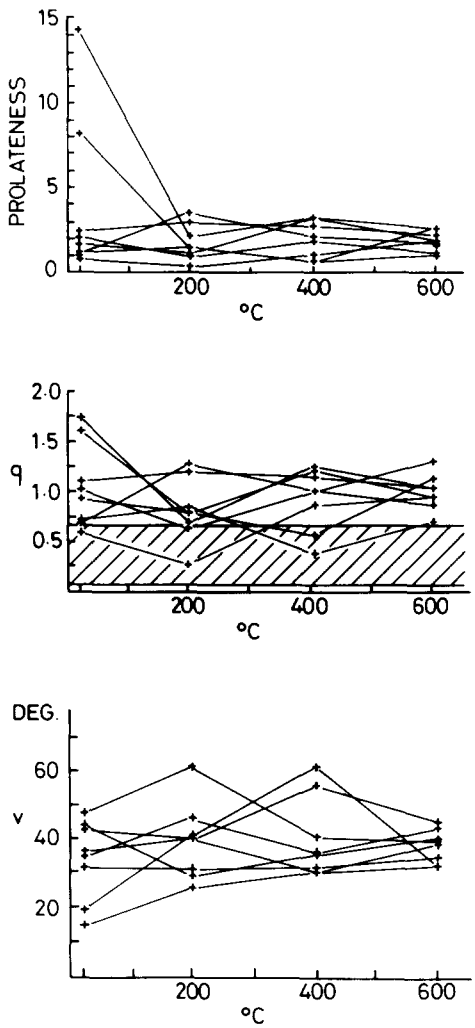


Fig.2. Magnetic fabrics as a function of temperature. The previous data on magnetic anisotropy (Abouzakhm and Tarling, 1975) is recalculated for (from top to bottom): the magnitude of sedimentary fabric, the sedimentary fabric parameter q, as used by Rees (1966), and the degree of prolateness. The diagram for q also shows the upper limit adopted by Rees (op. cit.) as indicative of primary sedimentary fabrics, indicating that the observed magnetic fabric is almost entirely secondary.

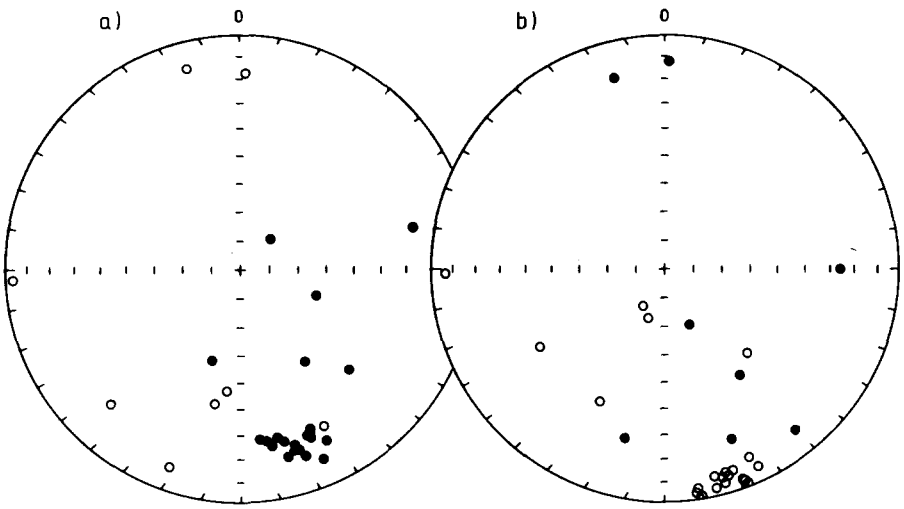


Fig.3. Directions of remanent magnetization for samples from the Garvellachs: (a) uncorrected for bedding tilt, and (b) corrected for tilt.

group, showed intermediate directions with steeper inclinations although correction for tilt of the bedding planes gives slightly lower inclinations (Fig.3).

One specimen from each of the 30 sites was submitted to incremental thermal demagnetization. The specimens showed variable degrees of stability, with all specimens from the high-intensity sites showing good stability, and some of those from the low-intensity sites showing poor stability (Table I). A further 30 specimens, again one from each site, were then submitted to progressive AF demagnetization. The specimens again showed variable degrees of stability (Table I) with a good agreement with the thermal pilot specimens. (Some typical examples are shown in Figs.4 and 5.) The results from both treatments yield essentially identical results for the high-stability specimens. These specimens showed little variation in intensity and direction with fields of up to 80 mT and thermal demagnetization indicated blocking temperatures above 500° C, possibly in excess of 600° C (Fig.4a, b, c). This, however, is difficult to evaluate since the higher temperature decrease in intensity was accompanied by an increase in low-field susceptibility (Fig.4c), indicating chemical changes affecting the ferromagnetic mineralogy at high temperatures. For the low-intensity, less-stable specimens, thermal treatment yielded generally greater directional changes (Fig.5) but some specimens showed a stable behaviour at moderate fields and temperatures, indicating the presence of a stable magnetization. Studies of the high coercivity—high blocking temperature portions of the spectrum were restricted because of the lower intensities of magnetization and the presence of chemical changes.

TABLE I

Site initial mean values for the Garvellach Islands

Site	N	Present horizontal		Palaeo horizontal		K	α_{95}	-10^{-3} J_{NRM} (mA/m) site means	Stability indices		Lithology
		Dec.	Inc.	Dec.	Inc.				Thermal	AF	
1	6	108.7	62.1	145.4	44.4	3	42.3	0.03	0.48	0.78	dolomitic limestone
2	6	145.8	50.7	159.0	24.2	6	28.6	0.18	1.29	0.63	dolomitic limestone
3	6	152.8	19.0	154.4	−7.7	1090	2.0	13.53	22.71	19.97	varvite
4	6	174.8	28.3	176.0	−2.3	7	28.2	27.09	29.00	25.56	varvite
5	6	170.7	27.5	172.3	−2.7	235	3.9	50.19	15.61	20.07	varvite
6	6	165.5	24.8	167.3	−4.7	584	2.8	39.28	16.63	14.68	varvite
7	6	162.2	19.8	163.4	−9.0	521	2.9	33.10	32.90	30.37	varvite
8	6	167.7	28.7	169.9	−1.1	394	3.4	38.69	7.70	18.99	varvite
9	6	156.9	23.7	159.4	−4.1	58	8.9	28.66	18.25	37.01	varvite
10	6	170.1	23.9	171.4	−6.1	555	2.8	32.33	33.00	46.16	varvite
11	6	156.4	13.4	146.3	−14.0	377	3.5	8.69	29.80	22.83	varvite
12	6	163.2	20.6	164.5	−8.4	1414	1.8	30.74	25.07	21.23	varvite
13	6	160.5	17.6	161.3	−10.8	269	4.1	17.50	10.89	43.58	varvite
14	7	156.7	24.2	159.4	−3.6	34	10.5	7.92	14.39	20.29	varvite
15	6	163.0	17.8	163.7	−11.1	16	17.4	8.21	8.76	33.70	varvite
16	5	152.0	−25.4	136.1	−49.3	16	19.8	0.18	2.19	5.23	varvite
17	6	345.2	−13.1	345.1	16.6	5	33.7	0.39	3.60	5.18	limestone
18	6	1.5	−17.7	1.3	12.0	2	62.4	0.84	1.56	1.13	limestone
19	6	200.0	−11.3	207.2	−37.3	3	50.5	0.08	0.79	2.26	limestone
20	4	224.2	−20.6	238.5	−36.7	3	72.1	0.06	0.71	2.11	limestone
21	6	163.2	23.5	163.9	−6.1	143	5.6	101.25	10.63	22.06	blue limestone
22	6	166.1	18.4	166.4	−10.4	528	2.9	131.03	30.19	29.10	blue limestone
23	6	186.7	−47.4	211.4	−75.1	268	4.1	5.09	18.72	25.90	pink limestone
24	6	155.8	27.8	157.3	−1.1	3	45.9	0.09	1.09	0.88	limestone
25	5	132.3	39.3	141.2	13.9	3	54.9	0.06	1.83	1.10	limestone
26	6	74.7	25.9	89.8	26.3	5	35.0	0.04	0.96	0.47	limestone
27	5	190.6	−41.4	198.6	−72.0	125	6.9	2.77	23.08	7.60	dolomitic mixtite
28	6	44.3	74.9	158.1	68.7	19	15.9	0.59	3.68	1.65	mixtite
29	6	266.7	−2.2	269.0	−6.2	2	66.7	0.05	0.56	2.49	mixtite
30	6	199.1	57.1	193.4	26.7	6	30.5	0.14	1.46	7.30	mixtite

The number of separately oriented samples (N) is the same as the number of specimens used, i.e. one specimen per sample. The stability indices for thermal and alternating field demagnetization of one pilot specimen per site are as defined by Tarling and Symons (1967).

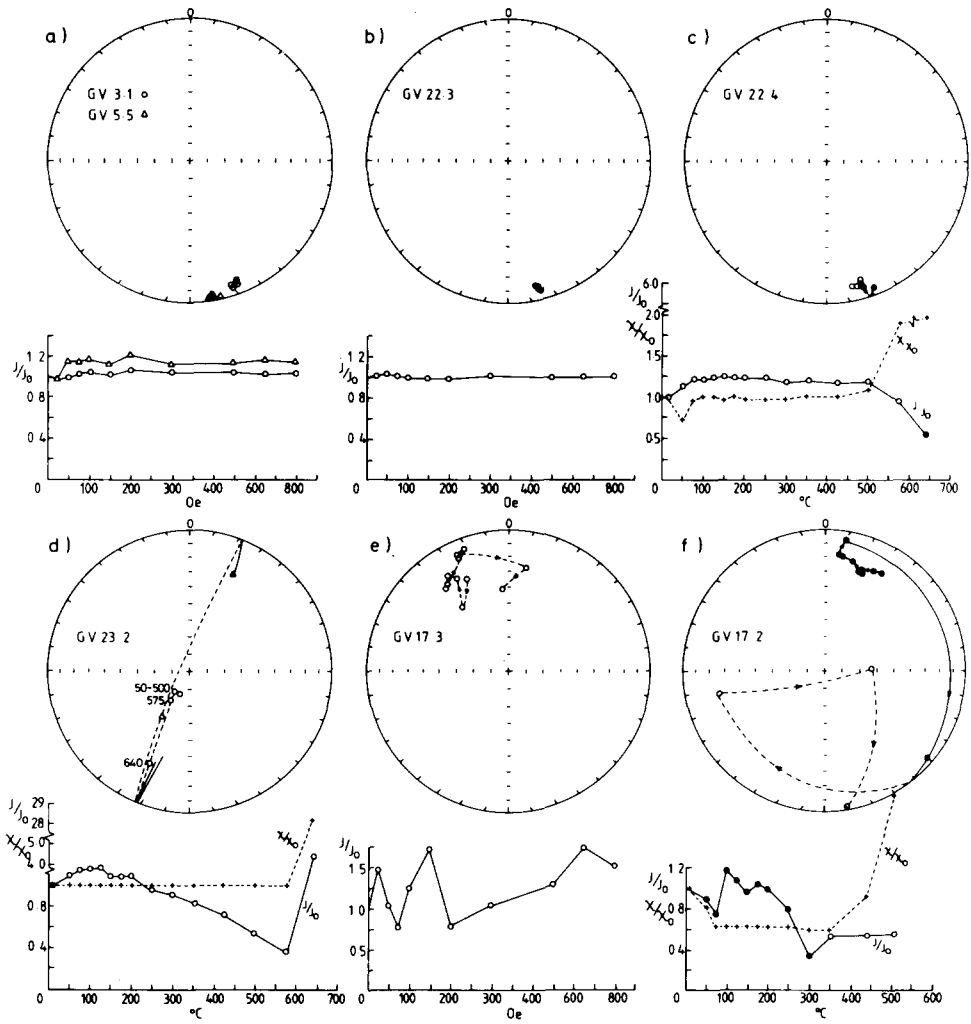


Fig.4. Alternating magnetic field and thermal demagnetization behaviour of specimens from the Garvellachs.

The analysis of weakly magnetic rocks presents more difficulties than that of strongly magnetic ones. One of the major problems lies in the measurement of the remanence. To ensure, as objectively as possible, that the data obtained were reliable, a special set of laboratory procedures was used. Specimens were measured several times, depending on the signal—noise ratio. The signal of the empty sample holder was measured at regular intervals during the measurements. These noise-level results showed that the noise level was effectively reduced by the increasing time of integration of the signal (Molyneux, 1971). However, a practical limit was observed beyond which the noise level started to fluctuate and increase again. This limit imposed a

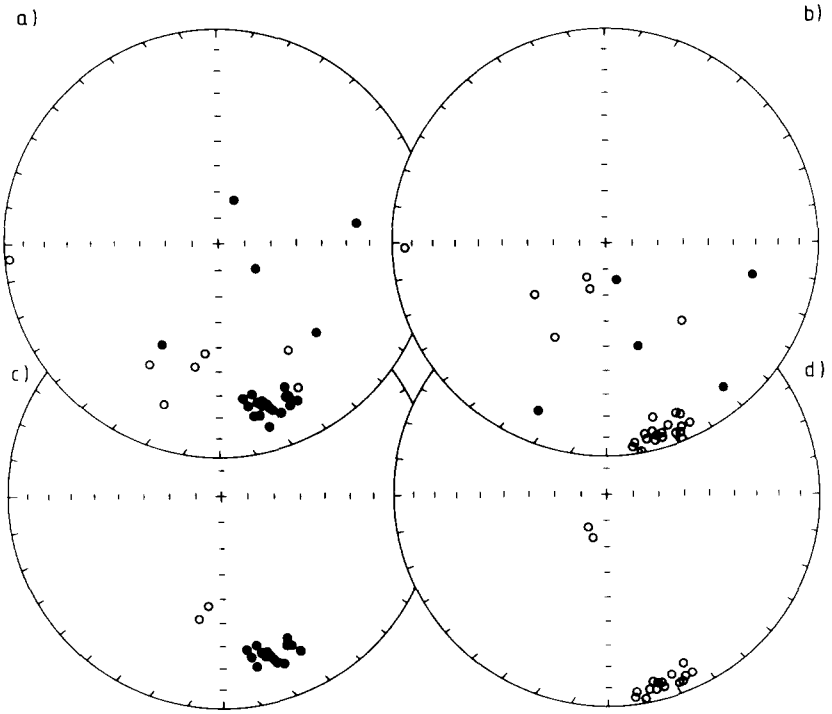


Fig.5. Partially demagnetized sample directions from the Garvellachs. All directions are shown in (a) and (b), before and after correction for tectonic tilt respectively. Sample directions for which the directions are thought to be adequately defined as shown in (c) and (d), before and after correction for tectonic tilt respectively.

practical constraint on the measurements and on the lower level of magnetization that could be measured with an adequate degree of reliability. Replicate observations obtained by measuring each specimen three times were treated as independent results and combined using Fisherian statistics. The internal dispersion of the directions about the mean, and the standard deviation of the intensities were used as control parameters to evaluate the results for each specimen. Large values of the internal dispersion parameter ($\alpha_{95} \geqslant 40°$) or of the intensity variation parameter (standard deviation $\geqslant 40\%$) were used as indicators of unreliable measurements, which were then repeated.

Two sites with high intensity and high negative inclinations (23 and 27) are thought to have been affected by heating from nearby Tertiary dykes as their directions correspond closely to those expected for the British Lower Tertiary. One of the pilot specimens from one of these sites, showed an angular movement at high temperatures (575° and 640°C) towards a southerly low-inclination direction (Fig.4d); thus giving some support to the idea of a secondary over-printing during the Tertiary. All remaining specimens from all sites were AF demagnetized at a 20 mT peak field (Table IV and Fig.5).

The results of the high-intensity sites (Fig.5) are well grouped about a low-inclination southerly direction and their corresponding site mean pole positions (Table IV) are shown in Fig.9. These results agree with the previous results obtained by Tarling (1974) which were assumed to be primary (Eocambrian) magnetizations.

ISLAY

Sampling on this island was limited to two localities and included varves and mixites (Fig.6). A total of 21 cores were drilled from several beds of varved siltstones exposed on the eastern side of Loch Allan (Grid ref. 676 423), and a further 18 cores were drilled from mixite beds near Port Askaig (Grid ref. 693 431). The intensity of the initial remanence showed that the samples from Loch Allan formed two distinct groups of respectively higher ($\geqslant 10^{-3}$ A/m) and lower intensity ($\leqslant 0.2 \times 10^{-3}$ A/m). A total of 15 specimens were submitted to incremental thermal demagnetization and 5 specimens to progressive alternating field (AF) demagnetization. Results from the low-intensity samples showed poor stability, indicating that their intensity was within the noise level of the instrument. The higher intensity samples showed good stability to both AF and thermal treatments (Fig.7a, b). The results from these stable samples showed predominantly normal, but some

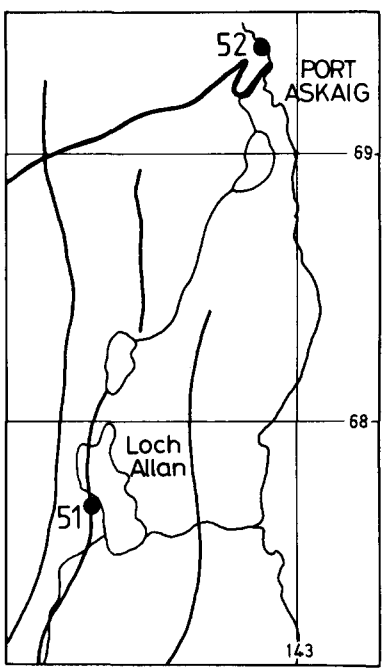


Fig.6. Sampling sites on Islay. The region within which British glacial deposits of this age have been round is shown in the left-hand corner, based on Spencer (1971).

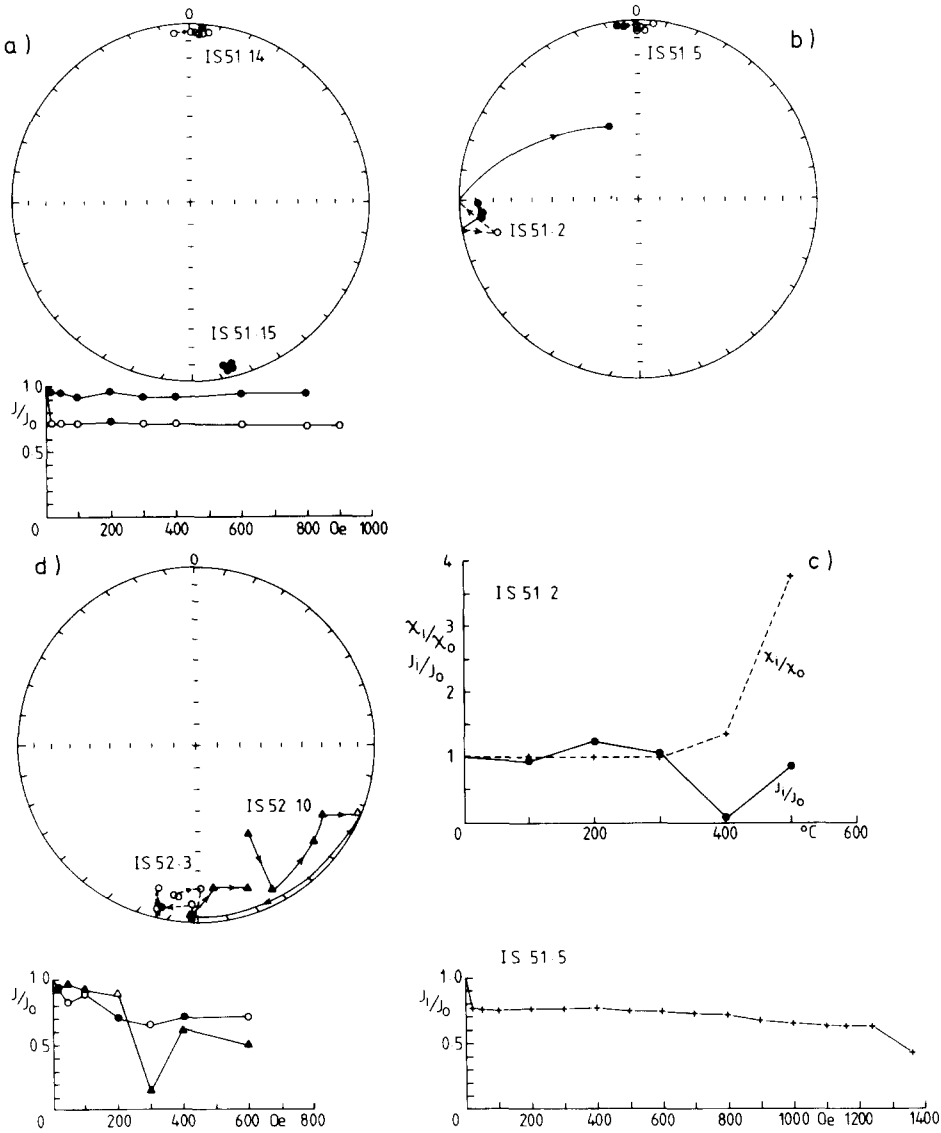


Fig. 7. Demagnetization behaviour of Islay samples. The alternating magnetic field demagnitization behaviour for typical examples of Islay Eocambrian sediments are shown in (a), (b) and (d) while the typical behaviour to thermal demagnetization is shown in (c) for which the change in susceptibility is also indicated.

reverse polarities with low inclinations, except for one sample which showed an intermediate direction (Fig.7c). All remaining samples were AF demagnetized at 20 mT. The results before and after demagnetization (Fig.8 and Table II) indicate two groups of normal and reverse directions of low inclinations.

The intensity of initial remanence of the mixite samples was relatively low ($\leqslant 0.4 \times 10^{-3}$ A/m) with greater scatter in directions. This, and the difficulties in determining the bedding plane, restricted the amount of work carried out on these samples. A total of 4 samples were submitted to incremental thermal demagnetization and 3 samples to progressive AF demagnetization (Table II and Fig.8c, d). The stability was poorer than that of the varves, but some samples showed directional changes about a direction of low inclination close to that of the varves (Fig.4d).

THE AGE OF REMANENCE AND PALAEOCLIMATIC CONSEQUENCES

In the Garvellachs, an identical direction of remanence with high stability has been isolated in both "glacial" and "low latitude" sediments, with very localised overprinting associated with Early Tertiary dykes. In Islay, a similar stable component has been isolated in "glacial" tillites although both normal and reversed components are present. The fundamental problem is thus to determine the age of this stable component in both islands. In the Garvellachs, correction for bedding slightly increases the precision of the mean direction, but the variation in bedding is small and hence such improvement is only indicative of a probable pre-folding origin. Unfortunately, the bedding in the mixtites in Islay could not be determined, so only the tillite site can be compared, before and after correction for tectonic tilt, with the Garvellach sites. Again, there is a decrease in the scatter after correction for bedding when both regions are compared. However, when only one site is involved, although involving 16 samples (19 specimens), such an improvement could again be fortuitous. The difference in inclination, 9°, remains the same between the two mean inclinations, although the declination value of the Islay site lies closer to the mean Garvallach declination value after correction for tilt, but this could indicate a small tectonic rotation about a vertical axis as both outcrops are well separated and almost certainly lie on different fault blocks. The indications are thus that the magnetization was acquired before the Caledonian tilting and metamorphism, but this cannot be considered finally proved, and hence the observed directions can only be effectively dated by comparison with data of similar age from the same tectonic block.

The Garvellach and Islay region clearly lies north of the Caledonian suture line and hence pre-Silurian data from England and Wales are not directly relevant and the position of Eire, relative to the Caledonian suture line, is uncertain, but probably also lies on a different pre-Caledonian plate, which also differs from that of Britain in view of the "geosynclinal" sediments of Lower Palaeozoic age in Wales. The only directly comparable data from the same plate must thus be from Scotland, unless the magnetization was acquired

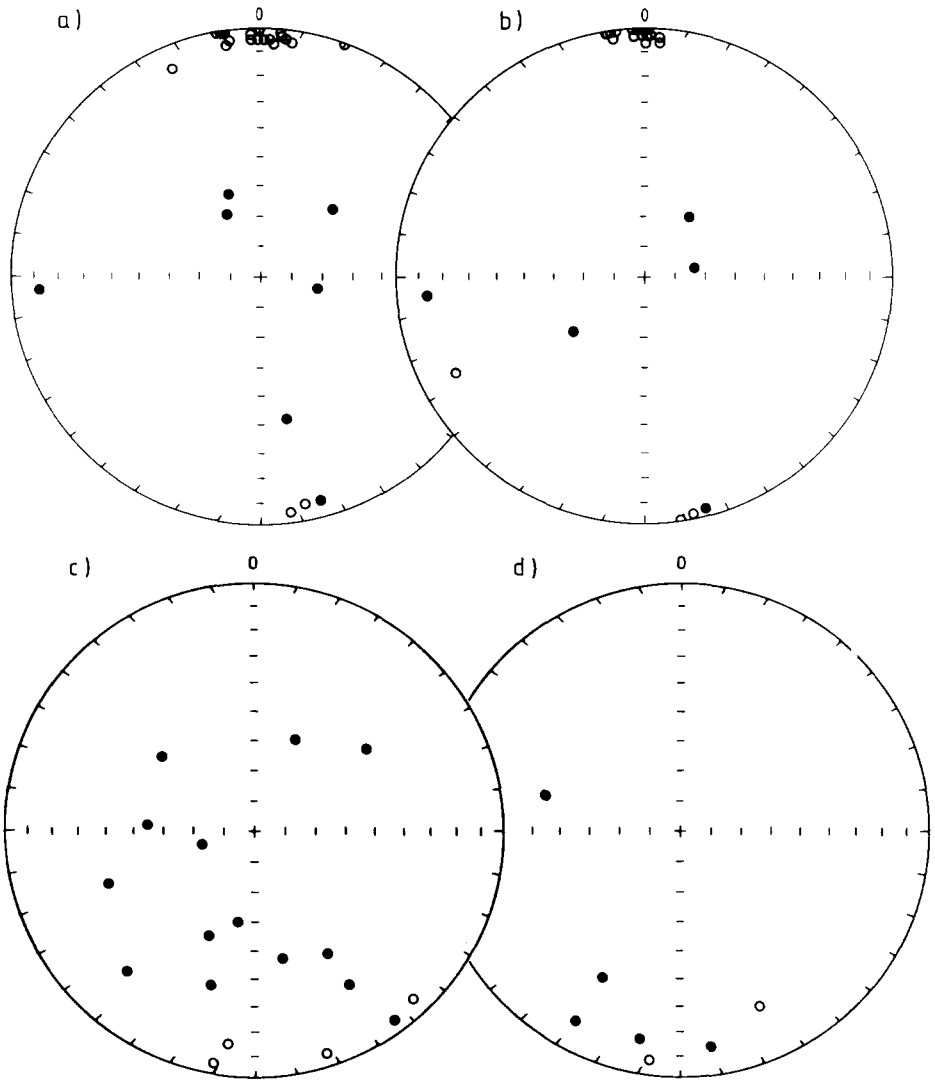


Fig.8. Specimen directions for Islay sites IS 51 and 52. (a) Before cleaning, and (b) after alternating field demagnetization. Both sterograms show directions after correction for bedding tilt for IS 51. The undemagnetized directions (c) and directions after alternating magnetic field demagnetization (d) are both shown uncorrected for bedding tilt as such a tilt was not possible to ascertain with sufficient reliability within the coarse sediments in site IS 52.

during the final stages of the Caledonian orogeny, i.e. during Silurian–Devonian times, as most areas north of the Hercynian Front would then have been in close contact, although it seems probable that different tectonic units would show some post-collision relative movements. Unfortunately, even the Devonian data from the British Isles are disputable (Tarling, 1979), although most postulated Devonian pole positions lie well to the east or south of the

TABLE II

Site mean values for Islay sites

Site	N	B	Present horizontal		Palaeo horizontal		K	α_{95}	J
			Dec.	Inc.	Dec.	Inc.			
Natural remanent magnetization									
IS 51.1	16	19	6.0	−15.0	0.8	−3.4	101	3.7	3.01
IS 51.2	5	6	319.0	48.3	4.7	72.4	3	48.6	0.19
IS 52	18	18	182.1	47.3	—	—	3	26.1	0.22
Cleaned remanence									
IS 51.1	16	19	3.2	−15.4	358.3	−2.4	253	2.3	3.01
IS 51.2	3	4	268.2	39.5	239.4	66.0	2	90.0	0.19
IS 52	7	7	197.8	16.0	—	—	4	36.0	0.22

The number of separately oriented samples (N) and number of specimens differ in this study. Some specimens were too weak to measure after partial demagnetization in 20 mT alternating magnetic fields. Site mean intensities (J) are calculated assuming a log-normal distribution ($\times 10^{-3}$ A/m).

observed northwestern Scottish poles (not illustrated). It seems reasonable to assume, therefore, that these Scottish directions were acquired in a pre-Devonian field. Comparison with the Eocambrian to Ordovician poles for the British Isles (Fig.9) is difficult because of the very high degree of scatter. Most of the Eocambrian—Cambrian poles for England and Wales (Table IV) do not, in fact, even fall within the northern Pacific, even when their polarities are inverted to obtain the poles nearest to this region. The fact that the Caerfai and Hartshill poles lie very close to the new Scottish poles cannot, therefore, be considered meaningful, particularly as these must have lain on different tectonic plates at that time. The Ordovician—Silurian results from England and Wales are quite well grouped in the central equatorial Pacific, apparently quite separate from the Garvellach—Islay poles, although showing some stringing towards them, possibly indicating the presence of some later components of magnetization in some of the Ordovician samples. The Irish Ordovician—Silurian poles fall within most of the northern Pacific and, as such, they could be considered to include the Garvellachs—Islay poles. Similarly, the Scottish Ordovician—Silurian poles are somewhat less scattered but they have a wide distribution that includes the Garvellach—Islay poles within their general grouping. The agreement or disagreement with the published pole values cannot be considered to demonstrate that the Garvellach—Islay pole positions are, or are not, actually different from the Ordovician poles of either England, Wales, Eire or Scotland but this only indicates how poorly these other poles are grouped. As in the case of the fold test, the statistics would suggest differences may be there, without being sufficiently conclusive as to be fully convincing.

TABLE III

Site mean values for the Garvellach Islands after partial demagnetization

Site	N	Present horizontal		Palaeo horizontal		K	α_{95}
		Dec.	Inc.	Dec.	Inc.		
1	6	160.5	75.5	157.1	75.1	7	28.0
2	5	145.1	30.4	151.2	5.3	9	26.6
3	6	161.0	23.3	162.9	−5.3	16	17.0
4	6	194.5	28.6	193.4	−2.0	6	31.3
5	6	161.2	27.6	172.8	−2.7	427	3.2
6	6	164.8	24.3	166.6	−5.0	551	2.9
7		163.0	19.6	164.0	−9.3	545	2.9
8	6	166.5	28.4	168.8	−1.2	316	3.8
9	6	156.8	23.7	159.3	−4.1	53	9.3
10	6	170.6	24.2	171.9	−6.0	570	2.3
11	6	156.5	13.4	154.6	−14.0	427	3.2
12	6	163.0	20.4	164.2	−8.6	1120	2.0
13	6	161.2	17.2	161.8	−11.3	387	3.4
14	7	156.9	24.1	159.5	−3.7	34	10.5
15	6	157.2	20.0	158.8	−7.7	132	5.9
16	5	158.0	−26.5	143.2	−52.5	29	14.5
17	6	155.4	19.2	156.2	−9.5	6	28.7
18	4	168.3	18.8	168.5	−11.1	12	27.5
19	6	199.1	−17.6	208.4	13.7	3	51.4
20	4	199.6	−59.1	270.8	76.6	2	90.0
21	6	161.8	23.4	162.7	−6.1	136	5.8
22	6	164.9	19.3	165.2	−10.4	574	2.8
23	6	186.7	−47.3	211.4	−74.0	232	4.4
24	6	130.9	34.9	139.0	11.0	3	42.7
25	5	140.0	31.6	145.2	5.6	3	56.2
26	5	74.9	31.8	93.7	31.1	6	34.3
27	5	189.9	−42.3	197.6	−73.1	157	6.1
28	6	34.0	74.7	161.9	71.0	41	10.6
29	6	273.1	8.4	269.0	6.2	2	72.6
30	4	208.8	43.4	202.6	4.3	18	22.5
mean $N = 15$[1]		162.0	20.7	—	—	104	3.7
		—	—	163.5	−6.7	159	3.0
mean $N = 21$[2]		162.0	24.0	—	—	31	5.8
		—	—	163.8	−4.1	33	5.5

As for Table II, the number of specimens is the same as the number of separately oriented cores (N). All specimens were subjected to 20 mT alternating magnetic field. The overall means were calculated only for those sites with an initial mean intensity of magnetization greater than 2.5×10^{-3} A/m (1) and then incorporating six weaker site mean values that lay very close to the main, higher intensity, group (2).

TABLE IV

Lower Palaeozoic palaeomagnetic poles for the British Isles

	Age	Lat (N)	Long (E)	Pole list no.
Eire (Ord—Sil)				
Killary	MO	9	214	11/80
Grangebeeth	O	22	160	15/139
Portrane	O	32	146	14/387
Portanf	O	32	146	15/138
Kildare	O	21	176	14/388 and 15/136
Connemara	∈—O	15	129	14/410
Glensaul	O	7	155	15/134
Glensaul	O	43	205	15/135
Mweelrea	LO	11	218	14/398
Contacts	US	42	248	15/130
England—Wales (Ord—Sil)				
Trefgarn	LO	5	172	14/402
Breidden Hill	MO	−1	197	15/133
Breidden Hill	MO	4	193	0/552
Eycott Group	LO	14	165	9/129
Borrowdaile	O	9	176	14/400
Borrowdaile	O	0	203	16/180
Builth Intrus	LO	2	182	14/390
Builth Volcs.	LO	16	179	14/399
Carrock Fell	LO	23	188	14/391
Llanvrin Lavas	MO	15	162	9/130
Arenig	LO	11	168	9/131
Snead	LS	−18	226	16/169
Caradoc	UO	23	178	16/172
Llandowyn US-	US—LD	7	130	15/224
Llandowyn	US—LD	48	176	15/225
Scotland				
Aberdeen Gabbros		11	189	14/401
Ballantrae Complex		12	207	
Byrne Hill		10	206	
Foyers		16	168	
Foyers		29	160	14/380
Bail Hill Volcs		33	168	
Canup	M∈—LD	29	160	0/306
Garabel-Fyne	US—LD	5	146	12/138
Arroclar	US—LD	8	144	12/139
ORS Lavas	US—LD	−2	147	14/362
ORS Lorne	US—LD	−2	141	14/377
ORS	US—LD	−2	141	14/380
Eocambrian—Cambrian (Eng—Wales)				
Caerfai	L∈	26	169	13/66
Harshill	L∈	18	165	5/83
Langagnd	∈?	19	223	

TABLE IV (continued)

	Age	Lat (N)	Long (E)	Pole list no.
EoCambrian "Glacials"				
Islay uncorrected varves		26	170	
Islay corrected varves		33	176	
Islay uncorrected mixtite		24	155	
Garb Eilach uncorrected		21	193	
Garb Eilach strung corrected		36	195	
All uncorrected		20	193	
Corrected		34	194	

Only three poles lying in the Pacific region, irrespective of polarity, are shown. The ages are ∈ = Cambrian, O = Ordovician, S = Silurian, D = Devonian; L, M, and U refer to lower, middle and upper.

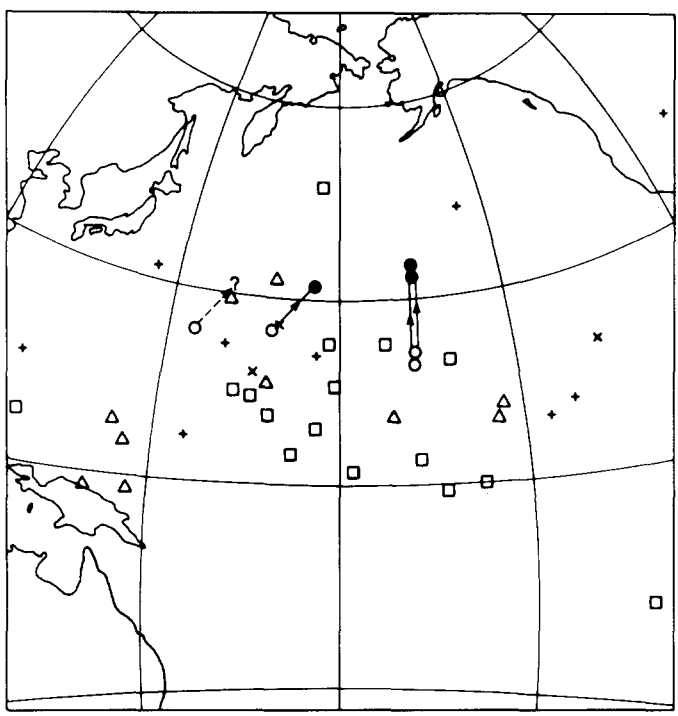


Fig. 9. Lower Palaeozoic palaeomagnetic poles in the Pacific Region. These poles are tabulated in Table III. Those from sites in Eire are shown as +, from Scottish sites as × for Eocambrian and △ for Orodivician—Silurian. English and Welsh Ordovician—Silurian sites are marked as □. The new Eocambrian plots are shown as ○ for uncorrected, and • for bedding corrected positions.

The age of the remanence in the Garvellachs and Islay region still cannot, therefore, be finally established. The indications are that the magnetization is likely to be primary, but an Ordovician—Silurian age cannot be ruled out. There is, however, very great stability and evidence for essentially single component remanence (i.e. excluding viscous remanence and locally demonstrable Tertiary overprinting). Clearly such materials are well worth collecting in other areas, although it is generally regarded that other areas in the British Isles show more extensive Caledonian metamorphism. Nonetheless, the establishment of a more precise "Caledonian" pole would, in itself, be of value. Meanwhile it seems reasonable to consider that the extant palaeomagnetic data are consistent with the palaeoclimatic indicators interbedded with the glacial varvites, suggesting a low latitude for the formation of both the glacial sediments and the limestones and carbonates which preceded their deposition. This would rule out explanations of the Eocambrian glaciation in terms of high palaeolatitudes (Crawford and Daily, 1971), with or without rapid polar shifts (Piper, 1973; McElhinny et al., 1974). The latter explanation would, in any case, be difficult to accept as constant cycling between the equator and latitude would then be necessary to account for the large number of glacial and interglacial episodes recorded in the British Eocambrian succession alone (Spencer, 1971). If the glacial and interglacial sediments were deposited in the same low latitudes, this does not rule out the possibility of glaciation induced by changes in the ecliptic (Williams, 1974, 1975), but this model seems itself to be unlikely as the deposits are thick and much more clearly segregated than would be expected if a six-monthly cycle were to be biased slightly one way and then reversed.

High altitude glaciation appears to be ruled out by the general palaeoenvironment, as the North Atlantic area was thought to be an area of shelf-seas with few high areas until the Caledonian orogeny commenced over 100 m.y. later. The sequence is clearly of marine deposits, and hence not at high altitudes. It is, of course, possible to question the glacial origin of these deposits as many of their features, particularly the tillites, are diagnostic of a seasonal climate and are not necessarily associated with freezing and thawing. Similarly many other features can have a different explanation. The key factor that makes a glacial origin virtually certain is the presence of dropstones. Although these do not show striae (which could, in any case, be attributed to non-glacial causes such as scree scratches), it is not possible to envisage a mechanism for their emplacement at a time when there was no vegetation to provide a floating mat from which stones could fall. The emplacement of these stones also rules out an origin by landslip, turbidity currents, etc. Thus while a non-glacial origin would solve many of the problems presented by these beds and their associated palaeomagnetic data, there seems no alternative to their explanation other than a world-wide glaciation as proposed by Harland (1964).

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